Engineering Mathematics (AHT-003)

Complete unit-wise notes - easy English, matched to the UTU syllabus.

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Unit 1 - Calculus I

What this unit covers (syllabus): limit, continuity and differentiability; Rolle's and mean-value theorems; Maclaurin/Taylor expansions (one and two variables); partial differentiation; maxima-minima and Lagrange multipliers.

Key ideas: A function is **continuous** at a point when its limit equals its value there; **differentiable** means the derivative exists (differentiable → continuous, not the reverse). **Rolle's theorem:** if f is continuous on [a,b], differentiable on (a,b), and f(a) = f(b), then some c has f'(c) = 0. **Mean value theorem (Lagrange):** f(b) - f(a) = f'(c)(b - a). **Taylor/Maclaurin** expand a function as an infinite polynomial - memorise the standard series: e^x, sin x, cos x, log(1+x), $(1+x)^k$. **Partial derivatives** differentiate with respect to one variable, treating the rest as constants. For **maxima-minima** in two variables: solve fx = 0, fy = 0, then check rt - s^2 (positive with r < 0 gives a maximum, positive with r > 0 a minimum). **Lagrange multipliers** handle constrained problems: solve ∇f = λ∇g.

Exam pointers: verify Rolle/LMVT for a given function (check the three conditions first - that is where marks hide), one Maclaurin expansion, and one maxima-minima of f(x,y) using the rt - s^2 test. Show every condition check explicitly.

Unit 2 - Calculus II

What this unit covers (syllabus): definite integrals and their properties, curve tracing, double and triple integrals, change of order and change of variables, Beta and Gamma functions.

Key ideas: Definite-integral properties save work: $\int_a^a f = \int_a^a f(a+b-x)$, and $\int_{-e}^{e} f = 2\int_0^e f$ for even f (0 for odd f). **Curve tracing**: check symmetry, origin, intercepts, asymptotes, tangents at origin before sketching - list the steps in the answer. **Double integrals** compute area/volume: integrate inner first, keeping the outer variable fixed. **Change of order**: redraw the region from the limits and rewrite with axes swapped - the sketch is everything. **Change of variables:** multiply by the Jacobian |J| (polar: dx dy = r dr dθ). **Beta/Gamma:** Γ(n) = (n-1)!, Γ(1/2) = √π, and B(m,n) = Γ(m)Γ(n)/Γ(m+n) - they convert hard integrals into one-line answers.

Exam pointers: one "evaluate the double integral" (often needing order change - draw the region!), one Beta/Gamma evaluation using Γ(1/2) = √π, and a curve-tracing question are the regulars.

Unit 3 - Calculus III

What this unit covers (syllabus): Jacobians, approximation of errors, applications of definite integrals (surface areas and volumes of revolution), centre of mass and centre of gravity.

Key ideas: The **Jacobian** J = ∂(u,v)/∂(x,y) is the determinant of partial derivatives; it measures how much a transformation stretches area. Key facts: J·J' = 1 (for inverse transformations) and if u, v are functionally dependent then J = 0. **Error approximation:** for z = f(x,y), the approximate error δz ≈ (∂f/∂x)δx + (∂f/∂y)δy - used for "percentage error" problems. **Applications of integrals:** area under/between curves, volume of revolution V = π∫y^2dx (about x-axis), and surface area of revolution. **Centre of mass/gravity:** x̄ = ∫x dm / ∫dm - for laminas use double integrals, for solids of revolution single integrals with density.

Exam pointers: compute a Jacobian (and test functional dependence), one error-approximation numerical (watch percentage-error wording), and a centre-of-gravity problem - the full worked example is in the notes under this unit.

Extra note - Centre of Gravity by Integration - made simple:

Centre of gravity (CG) = the point where the whole weight feels balanced. Like the middle point of a balance scale.

Formulas (for a lamina/plate):

- x̄ = (1/A) ∫∫ x dA , ȳ = (1/A) ∫∫ y dA
- where A = total area = ∫∫ dA

Steps: 1) Find the area A with a double integral. 2) Separately find ∫∫ x dA and ∫∫ y dA. 3) Divide each by A - done, you have (x̄, ȳ).

> *Think of tiny pizza slices. Add up each slice's "x contribution" (x dA), divide by the total area - you get the average x-position. That is the CG's x.*

Exam tip: if the shape is symmetric (like a quadrant of a circle or a rectangle), the CG lies on the line of symmetry - half the work done.

Unit 4 - Vector Calculus

What this unit covers (syllabus): vectors and their differentiation; gradient, divergence, curl; directional derivative; line, surface and volume integrals; Green's, Stokes' and Gauss divergence theorems.

Key ideas: A **scalar point function** gives a number at each point (temperature); a **vector point function** gives a vector (velocity field). **Gradient** $\nabla\varphi$ points in the direction of fastest increase; its magnitude is the maximum rate. **Directional derivative** = $\nabla\varphi \cdot$ (unit vector). **Divergence** $\nabla \cdot F$ measures outflow per volume (source or sink); **curl** $\nabla \times F$ measures rotation (curl $\nabla\varphi = 0$ always). Integrals: **line** (along a curve), **surface**, **volume**. The three big theorems convert between them: **Green's** (line integral around a plane curve ↔ double integral over the region), **Stokes'** (line integral around a closed curve ↔ surface integral of curl), **Gauss divergence** (surface integral of F ↔ volume integral of div F).

Exam pointers: compute grad/div/curl of a given field (easy, mechanical marks), one directional-derivative numerical, and one theorem verification - Gauss divergence is the most asked. Remember curl grad = 0 and div curl = 0; they appear as 2-mark proofs.

Unit 5 - Matrices

What this unit covers (syllabus): matrices and their types, rank, consistency of linear systems, eigen values and eigen vectors, Cayley-Hamilton theorem, diagonalisation.

Key ideas: Know the types: symmetric ($A = A^T$), skew-symmetric ($A = -A^T$, diagonal zeros), orthogonal ($AA^T = I$), identity, singular ($|A| = 0$). **Rank** = number of non-zero rows after reducing to echelon form. For a system $Ax = b$: consistent (unique solution) when rank(A) = rank([A|b]) = number of unknowns; infinitely many when both ranks are equal but smaller; **inconsistent** when rank(A) < rank([A|b]). **Eigen values** come from $|A - \lambda I| = 0$ (characteristic equation); for each λ, solve $(A - \lambda I)x = 0$ for the **eigen vector**. **Cayley-Hamilton:** every matrix satisfies its own characteristic equation - use it to find A^{-1} and higher powers. Properties: sum of eigen values = trace, product = determinant.

Exam pointers: the big repeated questions: find eigen values and vectors of a 3x3, verify Cayley-Hamilton and compute A^{-1} from it, and test a system for consistency using ranks. Trace/determinant cross-checks catch arithmetic slips - always do them.

Made by Anas. Free forever. Watch the linked videos on the site after reading each unit.